

2 (**short**) The semi-circular arch of radius R shown in Fig. 2 comprises three identical stiff, lightweight segments, pin-jointed at A, B and C, and built in at D. The arch is subject to a single applied force F acting vertically down at the centre of segment BC.

Derive a formula for the maximum bending moment in segment AB.

[10]

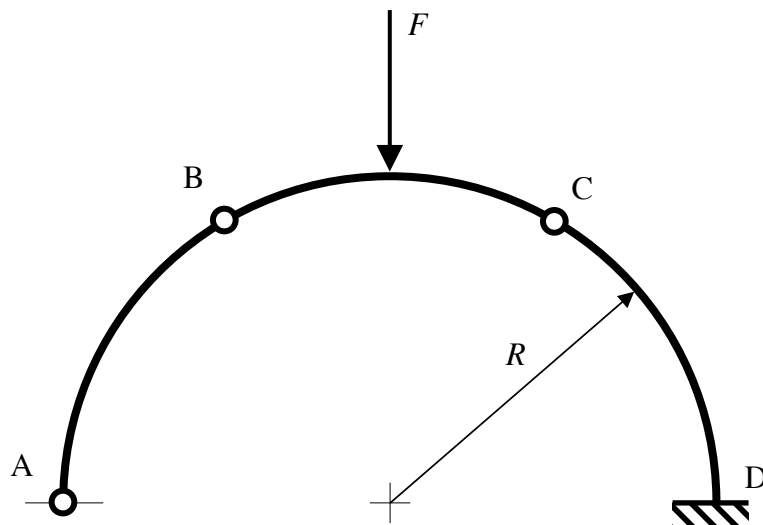


Fig. 2